

Choosing the Right Equipment: Balancing Equipment Cost and Project Sustainability for the Story Booth

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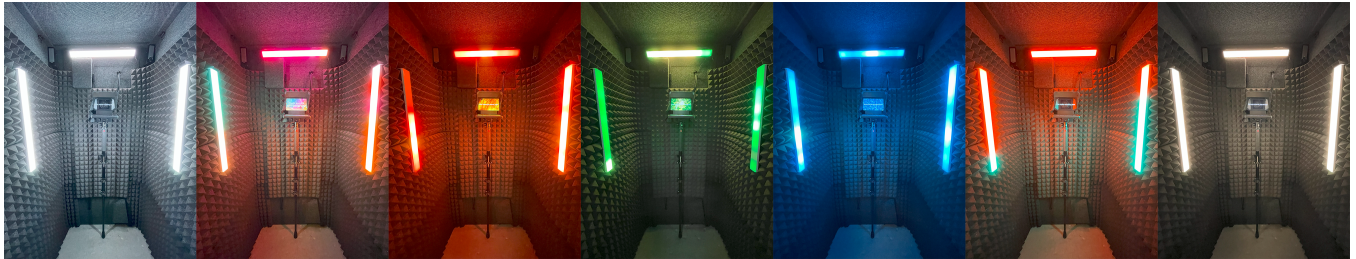


Figure 1: Images of the interior of the booth with various lighting

Abstract

In this workshop paper, we present the work choosing and implementing hardware for a techno-artistic project called “Story Booth”, which features a mobile recording booth outfitted with lighting and recording equipment. We describe a broad overview of the project and present our guiding themes behind hardware decisions of reuse, robustness and flexibility. For each theme we provide sample outcomes of our decision-making process. We conclude with a reflection on the benefits of our approach, which lead to higher initial investment in robust industrial equipment but has yielded an easy-to-use and easy-to-develop system with components that will continue providing value far beyond the initial project investment.

CCS Concepts

• **Hardware**; • **Human-centered computing**; • **Applied computing** → **Arts and humanities**;

Keywords

Lighting, DMX, Content Mapping, Arts, Soundbooth

ACM Reference Format:

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1 Project Background

The Story Booth is a portable volume providing a private space to record both audio and video. Questions are posed to participants via a teleprompter and computer-generated voice. Interior and exterior lighting are responsive to what is said by the participant. The interior lighting is intended to create a striking and dynamic appearance of the participant in the recording while the exterior is intended to spark interest and “draw in” future participants. There is intent to leverage Large Language Models (LLMs) for questions to provide novel user experiences. The goal of the booth is to operate nearly autonomously and to be taken to a variety of locations to collect the recordings of participants.

2 Hardware Description

The Story Booth is comprised of a wheeled booth, 87”x44”x44”[13], with a door on one face and sound attenuating foam across all interior walls. All surfaces of the booth are covered in carpet allowing for Velcro to attach. Inside is a teleprompter[7], camera[4], light bars[1], and a height adjustable stool. The exterior is covered in LED strips [9] and fabric. The computing hardware, video switcher[3], and LED controller [8] are stored on top of the box. A midi controller [11] is used for prototyping inputs. Power supplies and strips are attached to the back of the booth. Please see the BOM for a full description of the components.

3 Guiding Themes

We present the following themes guiding our decision-making process

3.1 Reuse

“Everything should come out at the end of the project and go into the next one”

The first key theme we wished to address in choosing equipment is promoting re-use of the equipment long term. To this end we focused on equipment that would not need extensive modifications, which features documentation provided by the manufacturer, and which requires a minimum of technical knowledge to operate.

We also adopted a modular approach when specifying components, looking for self-contained systems that could work together rather than trying to build one large system out of constituent parts. This allows for those self contained elements to be easily applied to future projects with no modification rather than trying to extract them from a bespoke installation.

Additionally, rather than permanently modifying any component for installation, we developed interstitial items to take advantage for instance, rather than applying adhesives or fasteners directly to the camera, tripod, or lights, we exclusively leveraged existing mounting points or fabricated mounting plates that could be modified to use existing attachment mechanisms like magnets.

3.2 Robustness

“We have to move it. We can’t deal with solder joints breaking every time we shift a component”

A key consideration with the Story Booth project, as with many artistic installations, is that the entire setup needs to be transported to a variety of locations. In particular, since we knew that we would want to transport the booth long distances to get a broad range of user group stories, we chose to identify components that would minimize risk of per-part failures.

The primary result of this was to leverage off-the-shelf components as much as possible, as these would have robust manufacturing built into their cost rather than taking additional time designing and fabricating on our side. In some cases this led to the decision to go with off the shelf components rather than wiring up custom systems. While “DIY” components could have been used and possibly provided more configuration possibilities, they would have required soldered connections and custom cut wiring which could fail or require repair which may not be possible in the field.

This further aided in our ability to iterate and improve upon the Story Booth. As design changes were requested we did not have to worry about the relative strength of a solder joint that had been repeatedly bent or low-cost mounting plates that had been repeatedly screwed into. Instead, we relied upon components that, from the manufacturers, were rated for this kind of stress and re-deployability.

3.3 Flexibility

“We want to spend our time prototyping and ideating, not soldering together LEDs”

Our final guiding theme was flexibility. By flexibility, we intended to select equipment and identify approaches that would allow agile development and the ability to quickly iterate through multiple prototypes without significant time or switchover cost.

Having much of our hardware as self-contained and battery powered, as described in section 3.1, we were able to rapidly relocate parts around the box to experiment with different lighting configurations and camera positions without the need to rewire components. Additionally, not having to rely on custom fabricated

cables allowed for the purchasing of a variety of lengths of DMX and HDMI cables to best suit the needs of the position of the components.

We also leveraged flexibility in our control systems. Utilizing DMX, an industry standard and widely supported protocol, as the underlying control protocol we were able to drive our lighting from various sources. From an off-the-shelf perspective we had utilities both from the manufacturer and from a third-party software that supports hardware input and output. This allows for flexible experimentation and iteration by users of a variety of skill levels and backgrounds. We were also able to implement a Python-driven DMX module which allows for fine-grained, programmatic control. By using this approach, we have also not been restricted to specific microcontrollers or required high power computing hardware.

4 Decision Making Outcomes

We present the following vignettes as examples of our decision-making process.

4.1 DMX-Controllable Lights

We chose the Aputure Infinibars [1] for the inside of the box since they use DMX control [2], an industry standard with existing usage in artistic endeavors, multiple presets which require no computer connected and flexible testing, and out-of-the-box feature an aesthetically pleasing enclosure with diffusion, mounting points, and controls.

4.2 Madmapper and MIDI control

While we have developed a robust set of Python modules for controlling our DMX equipment, all of our testing, evaluation, and prototyping has been conducted with Madmapper[10] and a MIDI keyboard [11]. This software+hardware combo allows us to literally play with the various components in our system and quickly demonstrate different potential approaches to scene setting and sequencing. While it is not suitable for the final deployment (which is intended to be autonomously deployed for extended periods of time), this setup allows a ton more flexibility and has provided an intuitive method for new team members to learn about the capabilities of the equipment.

4.3 Soundbooth and Velcro

The initial decision to leverage a sound booth was based on convenience: we already had a professional soundbooth on wheels[13] which provides a mobile, enclosed space that would be suitable for recording very high-quality audio. An unintended bonus to this choice was that the entire booth (inside and out) is covered with material that is compatible with hook-and-loop fasteners, meaning we can use the “sticky” side of Velcro to attach anything to the walls. This has allowed for rapid iterations of equipment choices, prototyping layouts, and attaching aesthetic/decorative items.

5 Discussion and Concluding Thoughts

Through the process of developing and deploying the Story Booth’s hardware, our key guiding themes served double-duty: they allowed not only for development of a solution that was approachable for those new to this kind of work as well as developers and designers

to join the team mid project, but also resulted in a system that required much less “people time” across the development process. While the decision to go with off-the-shelf and, in some cases, industrial equipment where new equipment needed to be procured lead to higher initial costs, we saw immense savings in the reduced personnel time needed for this project.

We also see benefit to future artistic endeavors in this initial outlay. Based on prior experience with projects that leverage “DIY” technology (e.g., custom LED strips, hobbyist microcontrollers, etc.), tend to not have much longevity. This arises from a combination of leveraging in-house manufacturing methods that are less robust and thus less resilient over time, leveraging in-house codebases that are often not maintained or well documented, and the tendency to use permanent or semi-permanent attachment methods for key technological components because they are “cheap enough.” The issue with components that break over time or code that becomes stale is obvious: once the initial funds for a project are no longer available, there is often no one to handle fixing, updating, and maintaining what was left behind and just as often no one familiar enough with initial design or coding decisions to be comfortable with maintenance resulting in the investment of both equipment and time going to waste. The danger of permanently building technology pieces into a project follows naturally from these two issues given, with no one to keep a project running, the ideal scenario is to recover salvageable goods and leverage them for the next project.

Based on our guiding themes, the equipment chosen for Story Booth makes strides to address these issues. The industrial lighting equipment is professionally manufactured with quality control and warranties, the code is kept minimal and over-documented, and our attachment methods are all reversible. While it is our goal for the Story Booth to be a long-standing installation, we also easily have the ability to recover every component (except consumable materials) and deploy them for future projects either by ourselves or by others.

It is worth noting that this approach is only made possible by the fact that we received an artistic grant from [INSERT FUNDING] which provided enough funds to cover a more expensive bill of materials. Our aim is not to simply equate “more spending” to “better projects” but rather to provide context and justification

for similar funding opportunities. For academics and researchers involved in artistic pursuits, positioning projects as a means to develop infrastructure, shared resources, and a generally available “toolbox” of equipment stands as a method to make any initial investments have impact far beyond any individual piece of work.

We also see great synergies between this approach to an artistic project and more traditional research at an institution. In particular, our experience with the Story Booth project has proven to be an excellent learning opportunity and has provided our unit with institutional knowledge of new products, approaches, and techniques to leverage on other projects.

Acknowledgments

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A Detail List of Materials and Components

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ITEM Purpose	Item Name
Booth	Whisper Room SE2000 Series
Teleprompter	Elgato Creator's Prompter
Camera	Blackmagic Design Pocket Cinema Camera 4K
Video Switcher	Blackmagic Design ATEM Mini Pro ISO HDMI Live Stream Switcher
Microphone	Shure MX202B/S
DMX Led Strip	CHAUVET Professional EPIX Flex 20 Outdoor-Rated Flexible LED Light Strip
DMX LED Controller	CHAUVET Professional ÉPIX Flex Drive Controller for ÉPIX Flex 20 Pixel LED Strip Light
LED Strip Power	CHAUVET Professional ÉPIX Flex Boost 24V Inline Booster
USB to DMX	ENTTEC DMX USB Pro
DMX Cables	Kopul DMX55P-005-S Studio Series 5-Pin DMX Cable
USB-C Harddrive	Samsung 4TB T7 Shield Portable SSD
XLR to mini XLR cable	Blackmagic Design Set of 2 Mini XLR to XLR Audio Cables for Video Assist 4K
DMX Light Bar	Aputure INFINIBAR PB6 RGB LED Light Panel (2')
Wired DMX Controller for Light Bar	Aputure Active USB-C to 5-Pin DMX In & Out Adapter for MT Pro, MC Pro & INFINIBARs